

Module 1 Review

Notations:

$\forall$: Mathematical notation for “For all”

$\exists$: Notation for “there exists”

$\in$: Mathematical notation for “Belongs to”

$\leq$: Mathematical notation for “less than or equal to”

$\mathbb{R}$: Set containing all real numbers

$\cap$: This is the notation for intersection of sets. Intersection of two sets results in a set that contains all the common elements between the two sets.

$\subseteq$: This notation can be read as “subset/equal to”. A set A is said to be a subset /equal to a set B if all the elements of a are contained in B.

$\not\subseteq$: Notation for “not a subset”

Mathematical Statement: *A mathematical statement or proposition or logical statement can be defined as grammatically correct sentence/sentences containing words, symbols and mathematical notations that has exactly one truth value which is either true or false.*

Variable proposition: *A variable proposition is a sentence/sentences the truth value of which depends on the value of a variable inside the sentence.*

Conjecture: *In mathematics, a conjecture is a **mathematical statement** which appears to be true, but has not been formally proven.*

Commonly used number sets in Mathematics

$\mathbb{R}$: Set of real numbers: Any value on the number line

$\mathbb{N}$: Set of Natural Numbers: Whole numbers from 1 upwards

Z: Integers: Whole numbers and negative whole numbers including 0

Q: Rational Numbers: Numbers you can make by dividing one integer by another (except dividing by 0)

I: Imaginary numbers: Numbers that when squared give negative results

Conjunction / And

If we have two statements: P, Q

To say that P and Q is true, both P needs to be True and Q needs to be True. Even if one of the statements is False, “P and Q” is False.

We use the symbol “ $\wedge$ ” to represent Conjunction/ And. This is called conjunction of P and Q.

Disjunction/ OR

If we have two statements: P, Q

To say that P or Q is True means that either P is True or Q is True (or both are True). So, if one of the statements is true, the entire statement has a Truth value of True. If both the statements are False, then “P or Q” is False.

We use the symbol “ $\vee$ ” to represent Disjunction/ Or. This is called disjunction of P and Q.

Negation/ NOT

If we have a statement P,

To say that Not P is True means that P is False. So when P is True, Not P is False and when P is False, Not P is True.

We use the symbol “ $\neg$ ” to represent Negation/Not. This is called the Negation of P.

Negation of “For all” is “There exists” and vice versa

P implies Q

Whenever P is True, Q is also true.

$P \Rightarrow Q$ or $P \rightarrow Q$

Such a situation where P is False, so, $P \Rightarrow Q$ is True, is called “**False Hypothesis**”.

Truth table example for implies and conjunction/disjunction

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$Q \Rightarrow P$
T	T	F	T	T	T	T
T	F	F	F	T	F	T
F	T	T	F	T	T	F
F	F	T	F	F	T	T

The conditional statement “ $P \Rightarrow Q$ ” and the statement “ $\neg P \vee Q$ ” have the same truth value.

Converse, Inverse and Contrapositive

Let P and Q be mathematical statements. Consider the “**original**” claim, $P \Rightarrow Q$.

Converse: $Q \Rightarrow P$

Inverse: $\neg P \Rightarrow \neg Q$

Contrapositive: $\neg Q \Rightarrow \neg P$

Logical Equivalence/Biconditional statements

If P and Q are two given statements, we say P is logically equivalent to Q if P and Q have the same truth value regardless of whether it's T or F .

We use the symbol " $\Leftrightarrow$ " to represent logical equivalence.

Necessary/sufficient conditions

General Rule of thumb:

In an "Implies" statement, the hypothesis is the sufficient condition, and the conclusion is the necessary condition.

De Morgan's Laws

$$\neg (P \wedge Q) \Leftrightarrow \neg P \vee \neg Q$$

And

$$\neg (P \vee Q) \Leftrightarrow \neg P \wedge \neg Q$$

Negation of conditional statements:

$$\neg (P \Rightarrow Q) \Leftrightarrow \neg (\neg P \vee Q)$$

$$\Leftrightarrow \neg (\neg P) \wedge \neg Q$$

$$\Leftrightarrow P \wedge \neg Q$$

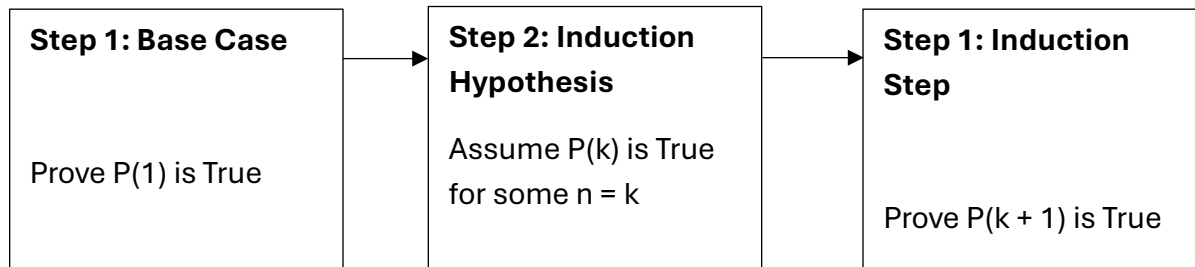
Practice: Proofs by semantics and Truth Tables

Principle of Mathematical Induction

Step 1: Verify if the statement is true for trivial cases ($n = 1$) i.e. check if $P(1)$ is true.

Step 2: Assume that the statement is true for $n = k$ for some $k \geq 1$ i.e. $P(k)$ is true.

Step 3: If the truth of $P(k)$ implies the truth of $P(k + 1)$, then the statement $P(n)$ is true for all $n \geq 1$.



Let's Practice

1. Write the English translations of the following statements and write their truth value:

- $\forall x \in \mathbb{R}. \exists y \in \mathbb{R}. x+y = 0.$
- $\exists y \in \mathbb{R}. \forall x \in \mathbb{R}. x+y = 0.$
- $\exists x \in \mathbb{R}. \exists y \in \mathbb{R}. x^2 + y^2 = -1$
- $\forall x \in \mathbb{R}. \exists y \in \mathbb{R}. x \leq y$
- $\exists y \in \mathbb{R}. \forall x \in \mathbb{R}. x \leq y$

2. Let P be a statement “13 is an odd number” and Q be the statement “4 is a prime number”. Write each of the following as English Statements and determine their Truth Value:

- $P \wedge Q$

- $P \vee Q$
- $\neg P \vee \neg Q$
- $P \wedge \neg Q$

3. Given conditional statement is True. Write the following as an implies statement with mathematical notations, then write the converse (using mathematical notations) and check if the converse is true. If it is true, write it as a biconditional statement (using notations):

If three points are collinear, they lie on the same line.

4. Let P represent a True Statement and Q and R represent a False Statement. Find the Truth Value of the following statement:

$$(P \Rightarrow \neg Q) \vee R$$

5. Let $x > 0$. Consider

$$Q: x^2 > x.$$

Determine whether each condition is **necessary**, **sufficient**, or **neither** for Q:

- (a) $x > 1$
- (b) $x > 2$
- (c) $x > 0$

6. Prove that $P \wedge Q \Rightarrow P$ using semantics and truth tables
7. Prove that $\neg(P \vee Q)$ is equivalent to $\neg P \wedge \neg Q$
8. Use PMI to prove the following for all $n \geq 1$

- $3n \geq 1 + 2n$
- $\sum_{k=1}^n (3k + 2) = \frac{3n(n+1)}{2} + 2n$

